

Aniruddha Kulkarni

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EDUCATION

NORTH CAROLINA STATE UNIVERSITY

Master of Computer Science

Relevant Coursework: Natural Language Processing, Neural Networks, Data Science, Software Engineering, Internet of Things

Raleigh, NC

May 2025

VISHWAKARMA INSTITUTE OF TECHNOLOGY

Bachelor of Technology in Computer Engineering

India

May 2023

SKILLS

Frameworks & Libraries: LangChain, FastAPI, PyTorch, TensorFlow, Hugging Face Transformers, Django REST Framework

Tools: Git, CI/CD (GitHub Actions), Docker, REST APIs, Vector Databases (Pinecone, FAISS), LLMs (GPT/LLama/Claude)

Specialization: Agentic AI (Agent Collaboration, Long-Term Memory, Agent Lifecycles), Multimodal Data Model Tuning, RAGs

Cloud Platforms: Microsoft Azure (Resource Groups, Azure OpenAI, Azure AI Search Service, Blob Storage, Embedding generation and vector retrieval pipelines), PaaS Deployments (Heroku, Vercel)

WORK EXPERIENCE

IBM

Project Intern

Bangalore, KA

Jul 2022 - Jan 2023

- Spearheaded an automated alert prioritization project across **IBM's Supply Chain Business Network**, analyzing **30K+ incidents across 10+** IBM Cloud servers using Python-powered data analysis to design a novel **three-tier priority policy**
- Designed and built **predictive machine learning models** using alert parameters like server type, alert resolution time to help set up automated **Python-based pipelines** to help resolve recurring faults and distinctly define short-lived "noise alerts"
- Streamlined alert notification policies by designing novel **Zabbix event triggers and PagerDuty's API** to ensure proper priority escalation for alerts, improving the alert-to-noise ratio for on-call IBM personnel by **87%**

Conexstra Technologies

Software Developer Intern

Pune, MH

Jul 2021 - Jan 2022

- Orchestrated the use of **AI and gamification technology** to construct environments with **100% accuracy** for a B2B SaaS platform, 'Praiso,' simulating road safety scenarios using **Python and CARLA**
- Utilized **image processing and data analysis** techniques to identify key indicators across **15+ scenarios** in improving driver safety to produce novel objects aimed at integrating and interacting with live simulation environments

Trident Labs

Senior Software Engineer

Pune, MH

Aug 2019 - May 2020

- Oversaw the development of one of the **earliest low-cost autonomous underwater carriers** out of India while **guiding** a team of **40+ individuals**
- Led the software development team developing novel algorithms for real-time underwater tasks involving **computer vision and deep learning on a Raspberry Pi**, achieving low latency/high accuracy solutions

PROJECTS

PDF-QA Assistant using Retrieval-Augmented Generation (RAG)

Updated Jan 2026

Built a production-grade RAG system enabling sub-second, grounded Q&A over large PDF corpora. Designed and integrated **Azure OpenAI** (text-embedding-ada-002, GPT-3.5-turbo) with **Azure AI Search** for embedding-based retrieval, using **Azure Blob Storage** for document ingestion and indexing. Implemented **vectorized search** over indexed PDFs supporting low-latency and cost-efficient cloud-native inference.

AI-Powered Multi-Agent Real-time Threat Assessment Platform ([Link](#))

Sep 2025

Built an AI-driven **crisis simulation platform** that transforms plain-English prompts into live, adaptive **multi-agent simulations** using **Next.js 14** and **Google Gemini 2.0 Flash**, modeling crises with actors, tasks, policies, and constraints to stress-test response strategies while improving safety, sustainability, and resource optimization.

AI-Powered Multi-Agent Software Engineering Platform (Venture picked by the NSF I-Corps)

Jul 2025

Built an agentic platform that plans, scaffolds, generates, reviews, and tests **production-grade software using LLMs**. Implemented dynamic **agent lifecycles**, **tech-stack-aware structuring** (React, Node.js, Python, SQL), long-term memory, and **inter-agent collaboration** using tools like LangChain, GPT-4, and vector databases.